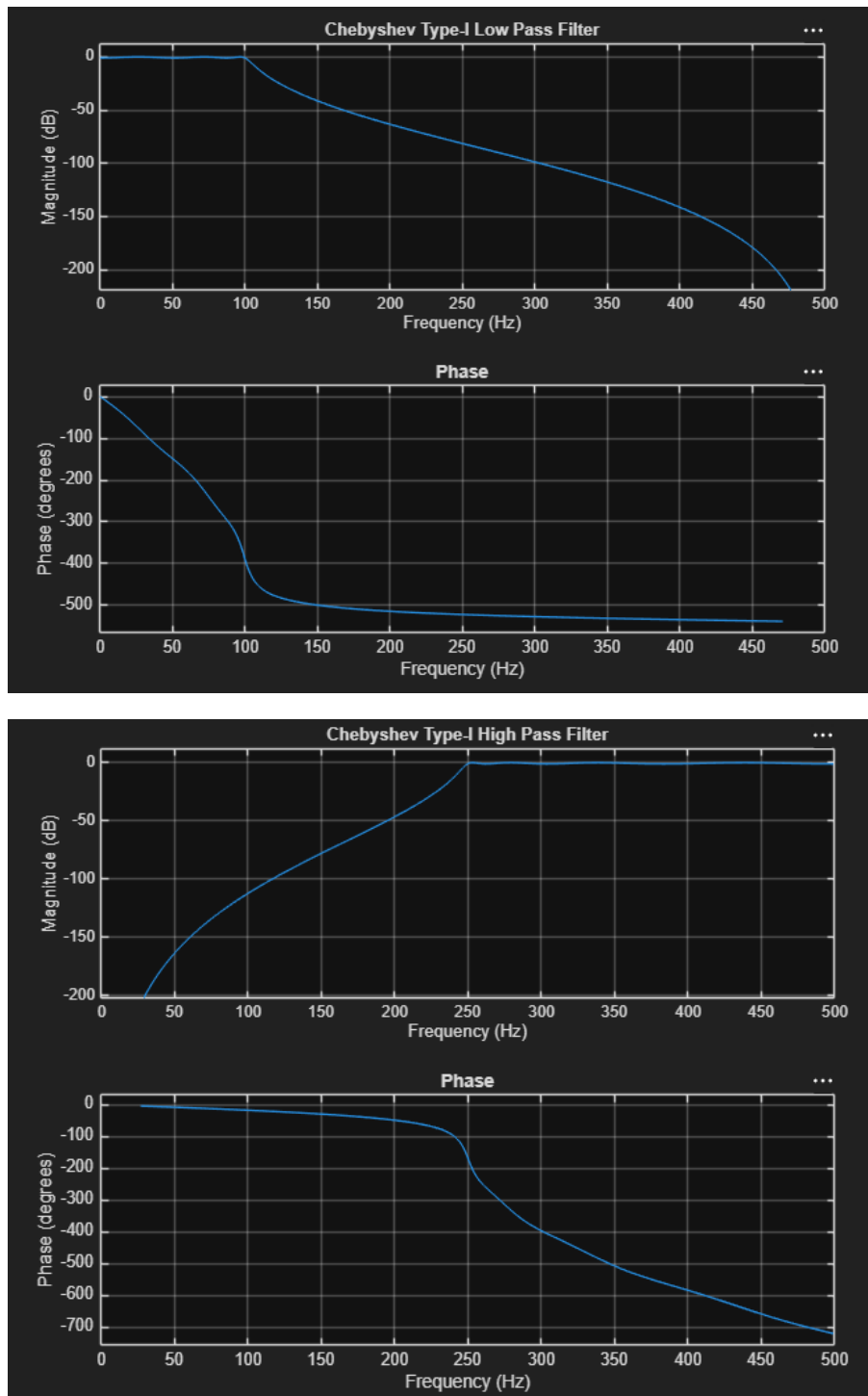
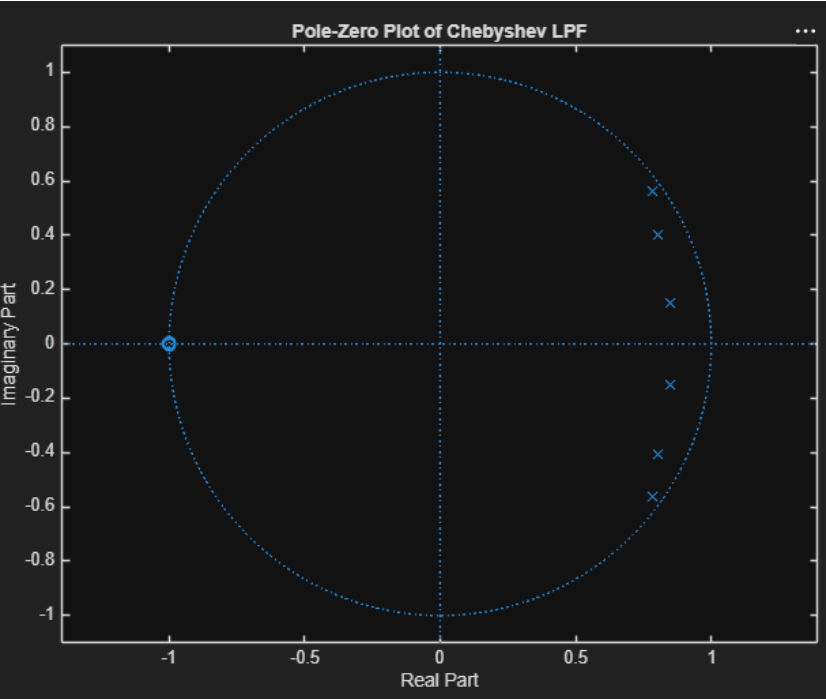
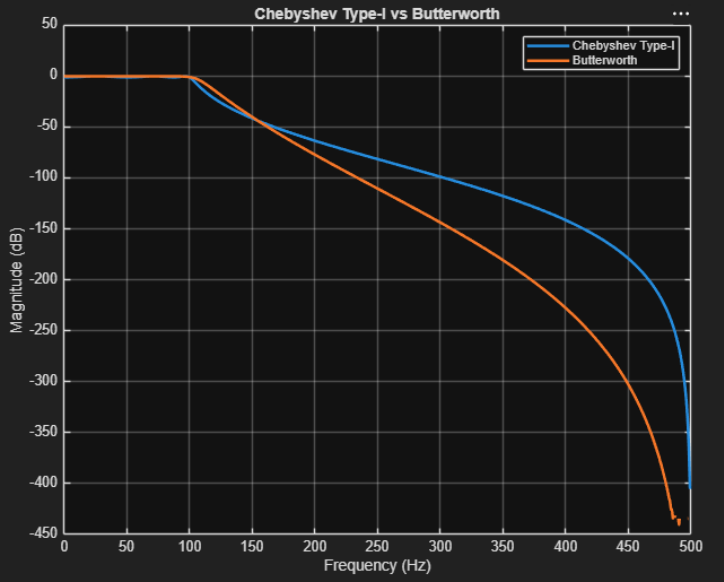
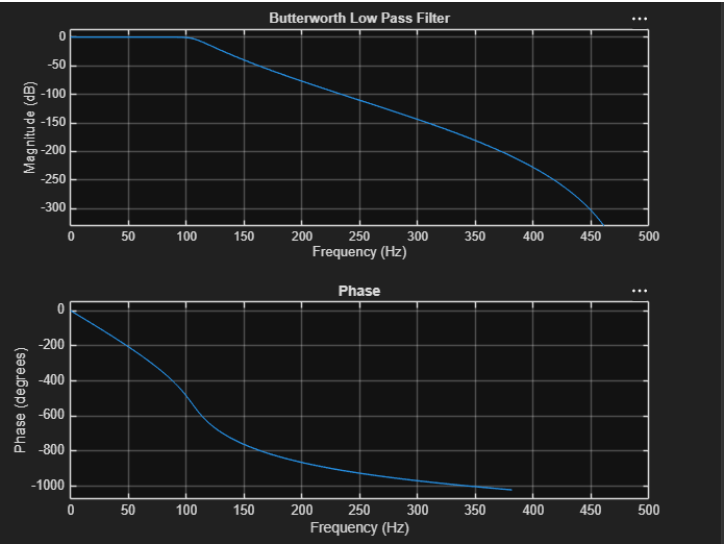


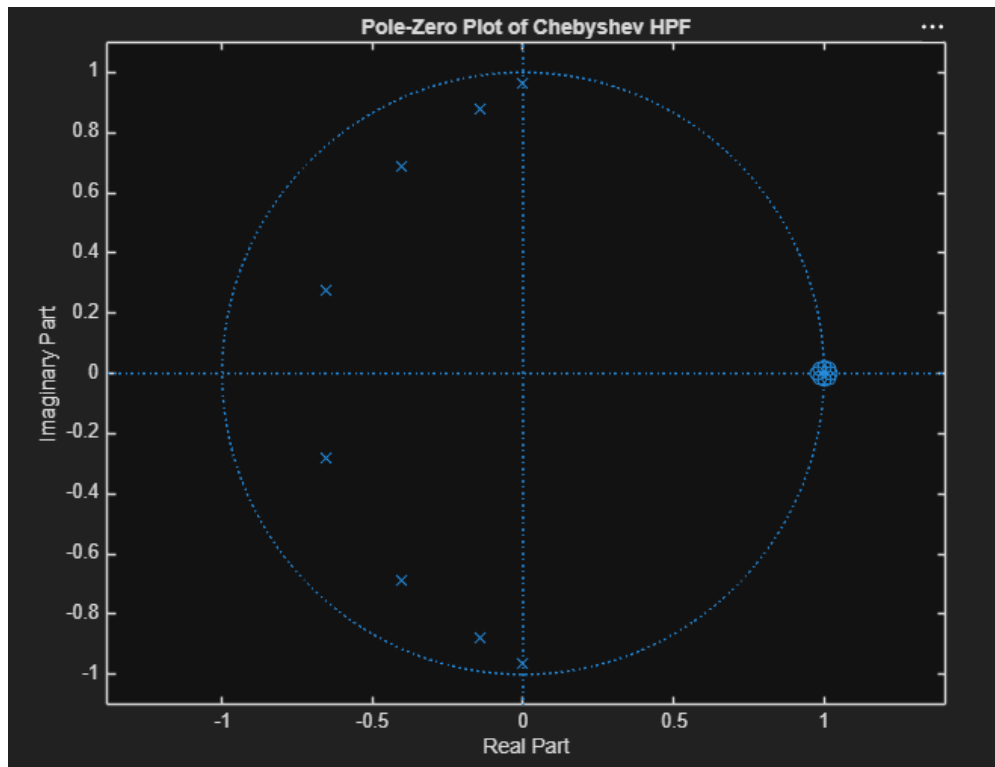
SIMULATION EXPERIMENT – 1

Design and Performance Analysis of Chebyshev Type-I IIR Low-Pass and High-Pass Filters Using MATLAB

OUTPUT:







***** CHEBYSHEV TYPE-I LOW PASS FILTER *****

Filter Order = 6

Numerator Coefficients (b):

1.0e-03 *

0.0464 0.2782 0.6956 0.9274 0.6956 0.2782 0.0464

Denominator Coefficients (a):

1.0000 -4.8694 10.3807 -12.3367 8.5969 -3.3268 0.5586

***** CHEBYSHEV TYPE-I HIGH PASS FILTER *****

Filter Order = 8

Numerator Coefficients (b):

0.0017 -0.0133 0.0465 -0.0929 0.1161 -0.0929 0.0465 -0.0133 0.0017

Denominator Coefficients (a):

1.0000 2.4048 4.5379 5.7984 5.7832 4.3883 2.5115 1.0019 0.2373

CHEBYSHEV TYPE-I LOW PASS FILTER

Filter Order = 6

Passband Ripple = 1.0 dB

Cutoff Frequency = 100.00 Hz

Usable Bandwidth = 100.00 Hz

CHEBYSHEV TYPE-I HIGH PASS FILTER

Filter Order = 8

Passband Ripple = 1.0 dB

Cutoff Frequency = 250.00 Hz

Usable Bandwidth = 250.00 Hz

BUTTERWORTH FILTER ORDER = 12

RESHEKESH B (19251213)